

override 2, 3, or even 100 methods? What if we wanted to create several different subclasses of `TTTBoard`? Not using inheritance would cause an explosion of `if-else` statements inside our methods and a large increase in our code's complexity. By using subclasses and overriding methods, we can better organize our code into separate classes to handle these different use cases.

The super() Function

A child class's overridden method is often similar to the parent class's method. Even though inheritance is a code reuse technique, overriding a method might cause you to rewrite the same code from the parent class's method as part of the child class's method. To prevent this duplicate code, the built-in `super()` function allows an overriding method to call the original method in the parent class.

For example, let's create a new class called `HintBoard` that subclasses `TTTBoard`. The new class overrides `getBoardStr()`, so after drawing the tic-tac-toe board, it also adds a hint if either X or O could win on their next move. This means that the `HintBoard` class's `getBoardStr()` method has to do all the same tasks that the `TTTBoard` class's `getBoardStr()` method does to draw the tic-tac-toe board. Instead of repeating the code to do this, we can use `super()` to call the `TTTBoard` class's `getBoardStr()` method from the `HintBoard` class's `getBoardStr()` method. Add the following to the end of your *tictactoe_oop.py* file:

```
class HintBoard(TTTBoard):
    def getBoardStr(self):
        """Return a text-representation of the board with hints."""
        ❶ boardStr = super().getBoardStr() # Call getBoardStr() in TTTBoard.

        xCanWin = False
        oCanWin = False
        ❷ originalSpaces = self._spaces # Backup _spaces.
        for space in ALL_SPACES: # Check each space:
            # Simulate X moving on this space:
            self._spaces = copy.copy(originalSpaces)
            if self._spaces[space] == BLANK:
                self._spaces[space] = X
            if self.isWinner(X):
                xCanWin = True
```